

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: ITA0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

*I S. Dharani Teja (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

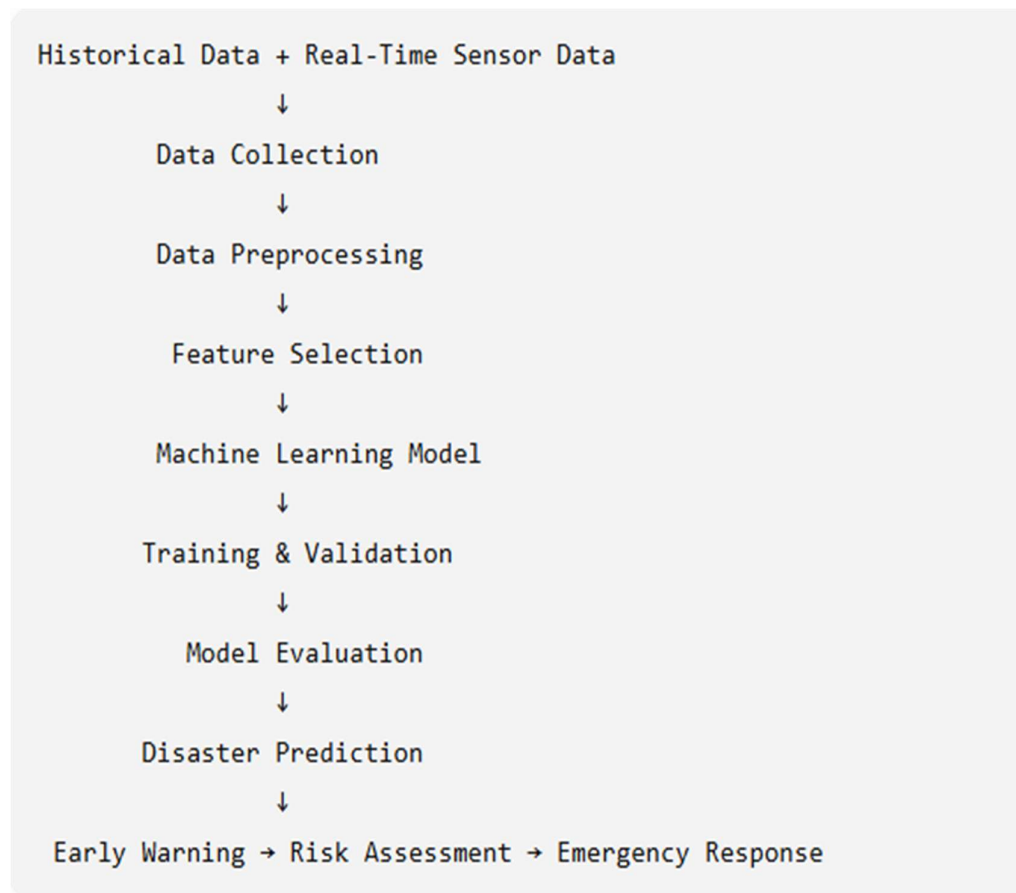
QUESTION 2 — Machine Learning System for Disaster Management

The question asks us to design a machine-learning system that predicts floods and earthquakes using historical and real-time data, justify the model, explain preprocessing and feature selection, and describe training, validation, evaluation, and practical use.

Answer

1. Proposed Machine Learning System

A disaster prediction system can combine **historical data** with **real-time sensor data** to predict the possibility of floods and earthquakes.



2. Data Collection

For **flood prediction**, the system can use:

- Rainfall
- River water level
- Soil moisture
- Temperature
- Humidity

- Drainage/water flow information

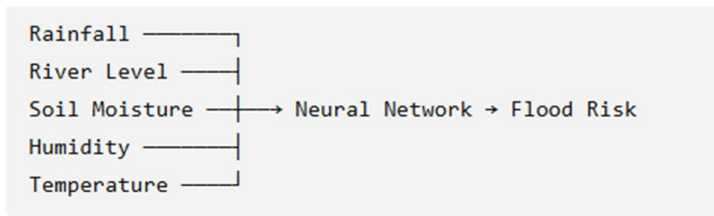
For **earthquake prediction**, relevant sensor information can include:

- Seismic activity
- Ground vibration
- Magnitude-related historical measurements
- Depth-related measurements
- Location of previous seismic events

Historical records are used for training, while real-time sensor data is used for current predictions.

3. Model Selection

A **neural network** is suitable because disaster prediction may involve several interacting input parameters.



A classification model can predict categories such as:

- Low Risk
- Medium Risk
- High Risk

For a larger and more complex system, different models can be evaluated and the model providing the best validation performance can be selected.

4. Preprocessing

The collected data may contain missing values, noise, and different scales.

The following preprocessing steps can be applied:

1. Remove or handle missing values.
2. Remove duplicate records.
3. Handle noisy sensor readings.
4. Normalize numerical features.
5. Convert categorical data into numerical form.

6. Divide data into training, validation, and testing sets.

5. Feature Selection

Features that contribute strongly to disaster prediction should be selected.

For example, flood prediction may prioritize:

Removing irrelevant features can:

- Reduce model complexity.
- Improve training speed.
- Reduce noise.
- Improve prediction performance.

6. Training and Validation

The historical dataset can be divided into:

Training Data → Train the model

Validation Data → Tune the model

Testing Data → Final performance evaluation

The model learns patterns from historical disasters and validates its performance on unseen data.

7. Evaluation

The model can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

For disaster management, **recall is particularly important**, because missing a real disaster can have serious consequences.

8. Practical Applications

The system can:

- Provide early warnings.
- Identify high-risk areas.
- Help authorities evacuate people.
- Support resource allocation.

- Improve emergency response planning.

Conclusion

A machine-learning-based disaster management system can combine historical and real-time data to identify disaster risks. A neural network can learn complex relationships between environmental and sensor parameters. Proper preprocessing, feature selection, validation, and evaluation can improve the reliability of the system.

QUESTION 4 — Genetic Algorithm for Candidate Solutions

The question provides five candidate solutions based on Speed, Accuracy, Fitness Value, and Suitability, and asks us to show how selection, crossover, and mutation improve the population.

Given Dataset				
Solution	Speed	Accuracy	Fitness	Suitable
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Step 1: Evaluate Fitness

The fitness values are already provided:

```
Solution 1 → 40
Solution 2 → 60
Solution 3 → 75
Solution 4 → 65
Solution 5 → 80
```

Higher fitness means a better candidate.

Therefore, ranking from best to worst:

80>75>65>60>40

So:

```
1st → Solution 5
2nd → Solution 3
3rd → Solution 4
4th → Solution 2
5th → Solution 1
```

Step 2: Selection

The best candidates are selected as parents.

Therefore:

SOLUTION 5 AND SOLUTION 3

are selected.

Solution 5 has fitness **80**, while Solution 3 has fitness **75**.

Why?

The Genetic Algorithm prefers individuals with higher fitness because they are closer to the desired solution.

Step 3: Crossover

The selected solutions are represented as chromosomes.

For example, a simplified encoding can be:

Solution 5 → Speed = Medium, Accuracy = High

Solution 3 → Speed = High, Accuracy = Medium

A crossover can exchange parts of these chromosomes.

For example:

Parent 1: Medium | High

Parent 2: High | Medium

↑

crossover

Possible offspring:

Offspring 1: Medium | Medium

Offspring 2: High | High

The second offspring combines:

- High speed
- High accuracy

This could produce a better candidate than the original solutions.

Step 4: Mutation

Mutation randomly changes one part of a chromosome.

For example:

Before mutation: Medium | Medium

After mutation: High | Medium

Mutation introduces new possibilities into the population.

It helps:

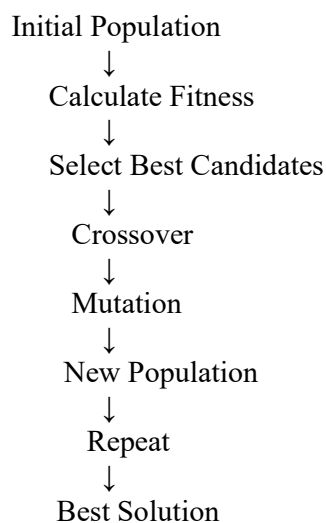
- Maintain diversity.
- Explore new solutions.
- Avoid premature convergence.
- Search different parts of the solution space.

Step 5: Best Solution

The original highest fitness is:

If crossover produces a candidate with both **High Speed and High Accuracy**, its fitness can potentially exceed the existing solutions, depending on the fitness function used.

Thus, the evolutionary process is:



Conclusion

Solution 5 is currently the best candidate with a fitness of **80**. Selection favors high-fitness individuals, crossover combines useful characteristics, and mutation introduces new variations. Repeating these operations allows the population to move toward an optimal solution.

QUESTION 5 — Neural Network for Disease Classification

This is one of the **strongest analytical questions** because the dataset actually provides **actual and predicted outputs**, allowing us to calculate TP, TN, FP, and FN.

Given Dataset		
Patient	Actual	Predicted
1	Yes	Yes
2	Yes	No
3	No	No
4	Yes	Yes
5	No	Yes

Step 1: Identify TP, TN, FP and FN

True Positive (TP)

Actual = Yes and Predicted = Yes.

Patients:

- . Patient 1
- . Patient 4

Therefore: TP=2

True Negative (TN)

Actual = No and Predicted = No.

Patient:

- Patient 3

- Therefore: $TN = 1$

False Positive (FP)

Actual = No but Predicted = Yes.

Patient:

- Patient 5

Therefore:

$$FP = 1$$

False Negative (FN)

Actual = Yes but Predicted = No.

Patient:

- Patient 2

Therefore:

$$FN = 1$$

Step 2: Confusion Matrix

	Predicted Yes	Predicted No
Actual Yes	TP = 2	FN = 1
Actual No	FP = 1	TN = 1

Step 3: Accuracy

$$\begin{aligned}
 \text{Accuracy} &= TP + TN / TP + TN + FP + FN \\
 &= 2 + 1 / 2 + 1 + 1 + 1 \\
 &= 3 / 5
 \end{aligned}$$

$$\text{Accuracy} = 60\%$$

Step 4: Precision

$$\text{Precision} = TP / TP + FP$$

$$= 2 / 2+1$$

$$\text{Precision} = 66.67\%$$

Step 5: Recall

$$\text{Recall} = \text{TP} / \text{TP} + \text{FN}$$

$$= 2 / 2+1$$

$$\text{Recall} = 66.67\%$$

Step 6: Specificity

$$\text{Specificity} = \text{TN} / \text{TN} + \text{FP}$$

$$\text{Specificity} = 1 / 1+1$$

$$\text{Specificity} = 50\%$$

Performance Analysis

The model has:

- Accuracy = **60%**
- Precision = **66.67%**
- Recall = **66.67%**
- Specificity = **50%**

Step 7: How preprocessing improves performance

Preprocessing can include:

- Handling missing values.
- Removing noisy data.
- Normalizing numerical features such as age, blood pressure, and glucose.
- Encoding categorical variables such as lifestyle risk.

Feature Selection

Important features such as:

should be retained if they contribute to prediction.

Removing irrelevant features can reduce noise and improve model performance.

Q7. Smart Traffic Management

The system predicts traffic congestion using vehicle density, time of day, weather, and road type.

System

Traffic Data



Preprocessing



Feature Engineering



ML Model



Training & Validation



Congestion Prediction



Real-Time Traffic Decision

- **Algorithm:** A classification model can predict **Low, Medium, or High congestion**. A neural network can be used when the relationships between features are complex.
- **Preprocessing:** Handle missing values, remove incorrect sensor readings, encode categorical data, and scale numerical features.
- **Feature engineering:** Create useful features such as **peak hour, day of week, and average vehicle density**.
- **Training:** Use historical traffic data to train the model and validation data to tune it.
- **Evaluation:** Use accuracy, precision, recall, F1-score, and confusion matrix.
- **Deployment:** Connect the trained model to real-time traffic sensors.

Conclusion: The predictions can help control traffic signals, identify congested roads, and suggest alternative routes.

Q12. Weather Forecasting

The system predicts rainfall using temperature, humidity, wind speed, and atmospheric pressure.

System

Weather Data



Preprocessing



Feature Engineering



ML Model



Training & Validation



Rainfall Prediction



Real-Time Alert

- **Algorithm:** A classification model can predict **Rain / No Rain**. A neural network is suitable for learning complex relationships between weather features.
- **Preprocessing:** Handle missing values, remove noise, and normalize numerical features.
- **Feature engineering:** Create features such as pressure change, average humidity, and wind-speed categories.
- **Training:** Train using historical weather data.
- **Validation:** Test the model on unseen validation data and tune its parameters.
- **Evaluation:** Use accuracy, precision, recall, F1-score, and confusion matrix.
- **Deployment:** Connect the model to real-time weather sensors for continuous predictions.

Conclusion: The system can provide real-time rainfall predictions and weather alerts.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand